



The Dawn, The Mountain and the Drone

SECURITY DEVICES

Using photography for security is not new. CCTV systems have been around from the late 1940s. With camera hardware improving rapidly, it is becoming cheaper and more viable for everyone to have a full-fledged security systems made up of hi-resolution cameras. The real excitement, however, is happening on the software side. With how good we've become in teaching AI to look at and recognize different things, we are getting closer to CSI's vision of being able to "ENHANCE" a picture 20 times to read licence plate numbers and identify faces. Big Brother conspiracy theorists will not like how smart CCTV systems are getting. They can accurately track faces from half a mile away, and even run facial recognition scans!

However, that's for more widespread security use. What about something much smaller, such as your car? Dash cams are also rising in popularity – at least in rest of the world. As they get cheaper, we can expect us Indians to also get them installed, given the amount of minor accidents that occur on a daily basis, and not to mention the weird stuff we see happening every day on Indian roads! Insurance companies will certainly find it easier to resolve cases using footage. However, there's another device with a camera that's already (often) placed on your dash. We're talking about your mobile, obviously, which is often on the dash in a holder giving you driving directions via Google maps. Whilst the screen is facing you, the phone camera is facing the road! With just a few minutes of tinkering around, you can repur-

pose your phone to become a dash cam. Look into it, and stay safe when driving!

SPACE PHOTOGRAPHY

Space photography really has to be one of the coolest things humankind has achieved. We mean, think about it! Capturing photons emitted by a star millions of years ago is just cool. Period. Ever since the aptly named 'Stargazer' went into orbit around the Earth in 1968, humans have been fascinated with capturing photos of the cosmos, free from the limitations posed by Earth on such a feat – stuff like light pollution and a pesky atmosphere that we unfortunately need to survive. But the breakthrough truly came through only with the launch of the Hubble in 1990. Here were never before seen photos of galaxies and stars in stunning detail. The Hubble has taken some memorable photos like those of the Pillars of Creation and the Monkey's Head Nebula and the James Webb Space Telescope



The Pillars of Creation is arguably the most famous space photo EVAH

is going to (hopefully) give us even more! The JWST packs a larger light catchment area that lets it peer back further in time than the HST, while also being farther away from Earth – HST orbits the Earth at about 570 kilometers out, while the JWST will orbit Earth a whopping 1.5 million kilometers away. But JWST isn't going to be a perfect replacement for HST because while the latter focuses more on the optical and ultraviolet capabilities, the former will focus on infrared. This means that the nature of photos taken will be radically different and it's going to perhaps show us a whole new side of Space we haven't seen before.

Overall, cameras and photography are going to find all new avenues to thrive in, because they are needed, and necessity is the mother of invention! 📷

Credit: Fernost | Wikipedia

When you can never have *two* much security